

Day 1 Evidence Workbooklet — Instructor Key

Visible Output and Stored State

Engineering practice → computational evidence → Python output/state → support route

Instructor key. Same layout as student copy; solutions filled in response boxes. Print format: duplex, left-stapled packet.

Name		Section		Date	
Score	____/33		Mastery check		secure / developing / needs support

Your team is doing a first-day hallway cart check. A small wheeled cart is rolled along a taped 12-meter path. One teammate times the run with a stopwatch. Another teammate enters the values in a Python notebook so the team can decide what should go into the lab record.

The problem today is not advanced physics. The problem is evidence: what did the notebook store, what did it visibly show, and which value is safe to write in the record?

Engineering task	Build a reliable evidence trail for two hallway cart trials before a result is copied into a lab record.
Computational move	Separate measurement inputs, stored variables, printed output, notebook display, and later overwrites.
Python focus	assignment, arithmetic expression, variable overwrite, <code>print</code> , final-expression notebook display, state after each cell.
Evidence to keep	exact visible lines, displayed values, stored values after each cell, a corrected lab note, and one non-mutating check.

Day 1 evidence rules

1. Assignment stores or overwrites a value. Example: `speed_mps = distance_m / time_s`.
2. `print(...)` creates visible output but does not by itself change a variable.
3. In a notebook, a bare expression at the end of a cell can display a value. Displaying a value is not the same as storing it.
4. A later assignment can overwrite an earlier stored value. Your trace must say what is stored *after each cell*, not just what looked important on the screen.

1 Read the notebook as an evidence system

recognize

predict

Use the scenario above to identify what kind of evidence each line can create. Do not calculate everything yet.

```

1 # Cell 1: first run on the hallway path
2 trial_id = "cart-A"
3 distance_m = 12.0
4 time_s = 3.0
5 speed_mps = distance_m / time_s
6 print("trial", trial_id)
7 print("speed", speed_mps)
8
9 # Cell 2: second run, same path, new time
10 trial_id = "cart-B"
11 time_s = 2.4
12 speed_mps = distance_m / time_s
13 print("trial", trial_id)
14 speed_mps + 0.2
15
16 # Cell 3: keep the adjustment separate
17 adjusted_speed = speed_mps + 0.2
18 print("stored", speed_mps)
19 print("adjusted", adjusted_speed)
20
21 # Cell 4: decide to record the adjusted value
22 speed_mps = adjusted_speed
23 print("recorded", speed_mps)

```

Pts.	Prompt	My evidence
1	Which variable names store measurement inputs rather than calculated results?	Key: distance_m and time_s. trial_id is a run label.
1	Which line first calculates a speed from the measured distance and time?	Key: speed_mps = distance_m / time_s
1	Which line in Cell 2 displays a candidate adjusted value without storing that adjustment?	Key: speed_mps + 0.2; no assignment stores it.
1	Which line in Cell 4 overwrites the stored value of speed_mps?	Key: speed_mps = adjusted_speed

2 Trace stored state after each cell

trace predict

Complete the state table. If a variable does not exist yet, write **not defined**. The first row has hints; the later rows require you to track overwrites.

Checkpoint	trial_id	time_s	speed_mps	adjusted_speed	Reason for any change
After Cell 1	cart-A	3.0	Key: 4.0	not defined	Key: Initial measured speed: 12.0 / 3.0.
After Cell 2	Key: cart-B	Key: 2.4	Key: 5.0	Key: not defined	Key: Trial/time overwritten; speed recalculated. Final expression displays 5.2 but does not store it.
After Cell 3	Key: cart-B	Key: 2.4	Key: 5.0	Key: 5.2	Key: Adjustment stored separately; speed_mps remains measured speed.
After Cell 4	Key: cart-B	Key: 2.4	Key: 5.2	Key: 5.2	Key: speed_mps = adjusted_speed overwrites the speed variable.

Pts.	Prompt	My response
2	At what checkpoint does speed_mps first become 5.0? Explain the measurement change that caused it.	Key: After Cell 2; time_s changes to 2.4, so $12.0 / 2.4 = 5.0$.
2	At what checkpoint does speed_mps first become 5.2? Explain what line caused the overwrite.	Key: After Cell 4; speed_mps = adjusted_speed .

3 Separate printed output from notebook display

predict

trace

Visible evidence can come from printed output or from a final notebook expression. Complete both columns. If a cell has no notebook display, write **none**.

Cell	Printed output lines from <code>print(...)</code>	Notebook display from final bare expression
Cell 1	Key: trial cart-A; speed 4.0	Key: none
Cell 2	Key: trial cart-B	Key: 5.2
Cell 3	Key: stored 5.0; adjusted 5.2	Key: none
Cell 4	Key: recorded 5.2	Key: none

Common trap to check

A displayed value can look official because it appears on the screen. But a lab record should say whether the value was printed, displayed as a candidate calculation, or stored as the current value after an assignment.

Pts.	Prompt	My response
2	In Cell 2, why is the displayed 5.2 not yet the stored value of <code>speed_mps</code> ?	Key: It is a bare expression result. No assignment stores it, so <code>speed_mps</code> remains 5.0.
2	In Cell 3, why is it useful to print both <code>stored</code> and <code>adjusted</code> ?	Key: It separates the current measured value from the adjusted candidate before overwrite.

4 Debug the lab record

debug explain test

A teammate writes this lab note after Cell 2:

Lab note to debug

Trial cart-B speed was recorded as 5.2 m/s because the notebook showed 5.2.

Pts.	Prompt	My response
2	What is accurate in the teammate's note?	Key: The notebook did show 5.2 after Cell 2.
2	What is inaccurate or incomplete in the teammate's note?	Key: After Cell 2, 5.2 was only displayed. It was not yet stored/recorded in <code>speed_mps</code> ; stored <code>speed_mps</code> was 5.0.
2	Rewrite the note so it separates measured speed, displayed candidate adjustment, and the value actually recorded after Cell 4.	Key: After Cell 2, cart-B measured speed was stored as 5.0 m/s and 5.2 was only displayed as an adjustment candidate. After Cell 4, <code>speed_mps</code> stored 5.2 m/s.

Pts.	Prompt	My response
1	Write one line you could run after Cell 2 to check the stored value of <code>speed_mps</code> without changing it.	Key: <code>print(speed_mps)</code> or bare <code>speed_mps</code> .
1	Write one line you could run after Cell 3 to check whether <code>adjusted_speed</code> exists.	Key: <code>print(adjusted_speed)</code> or bare <code>adjusted_speed</code> .
2	Before calculating speed, what measurement condition should the team check so the result is physically meaningful? Write it in plain language.	Key: Time must be positive/nonzero and correspond to the measured 12-meter path.

5 Compare two possible edits

[debug](#)[implement](#)[explain](#)

The team considers two ways to apply the 0.2 m/s adjustment. Both can show 5.2; they do not leave the same evidence trail.

Version A: keep candidate separate

```
adjusted_speed = speed_mps + 0.2
print("stored", speed_mps)
print("adjusted", adjusted_speed)
```

Version B: overwrite immediately

```
speed_mps = speed_mps + 0.2
print("stored", speed_mps)
print("adjusted", speed_mps)
```

Pts.	Prompt	My response
1	In Version A, what is stored in <code>speed_mps</code> after the code runs?	Key: 5.0.
1	In Version A, what is stored in <code>adjusted_speed</code> after the code runs?	Key: 5.2.
1	In Version B, what is stored in <code>speed_mps</code> after the code runs?	Key: 5.2; immediate overwrite.
2	Which version gives a better audit trail for a lab record? Explain your choice.	Key: Version A, because measured speed and adjusted candidate remain separate until the record decision.
2	If the goal is to record the adjusted value only after review, write one sentence explaining why Cell 4 is a safer place to overwrite <code>speed_mps</code> .	Key: Cell 4 overwrites only after both measured and adjusted values have been made visible.

6 Mastery check and support route

[transfer](#)[explain](#)

Use your own evidence from the workbooklet. Do not write a generic answer.

Pts.	Prompt	My response
2	What is one example from this notebook where the screen showed a value that was not yet the stored value that would be recorded?	Key: Cell 2 displayed 5.2, but <code>speed_mps</code> was still 5.0.
2	In one or two sentences, explain why engineers should preserve both visible output and stored-state evidence during early computational checks.	Key: Visible output shows what appeared on screen; stored-state evidence shows what the program will reuse or record. Keeping both prevents a displayed candidate from being mistaken for an approved/current result.

My mastery check

Mark the strongest statement that is true after this workbooklet.

☐	Secure: I can trace stored state and visible output separately, including a displayed value that is not stored yet.
☐	Developing: I can calculate some values, but I still need support separating display, print, assignment, and overwrite.
☐	Needs support: I need a smaller example before I can explain what Python stored after each cell.

My next support move

My issue is mostly: setup / Python syntax / tracing / arithmetic / notebook display / confidence / time.

The first place I got uncertain was: _____

My next small action is: ask a partner / ask instructor / rerun a cell / rebuild the trace table / try the recovery version.

Workbooklet target: I can build an evidence trail that separates measurement inputs, stored variable state, printed output, notebook display, and a later overwrite before a result enters an engineering record.

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